

GRNsight, automatically lays out a gene regulatory network (GRN) graph, coloring the edges based on regulatory weights, and the nodes based on experimental or simulated expression data

1. Network

- Demo files are provided.
- Can import Excel, SIF, or GraphML files.
- Can construct a network from SGD regulation or PPI data stored in a backend database.

2. Layout

- Users can select force graph or grid layout.
- Force graph parameter sliders:
 - Link distance determines the minimum distance between nodes.
 - Nodes have a charge which repels or attracts other nodes.

3. Node

- Coloring can be based on user-uploaded expression data or from published expression datasets stored in a backend database.
- Datasets available:
 - Apweiler et al. 2012, GSE33098
 - Barreto et al. 2012, GSE24712
 - Dahlquist et al. 2018, GSE83656
 - Kitagawa et al. 2002, GSE9336
 - Thorsen et al. 2007, GSE6068

4. Edge

- Edge coloring can be toggled on and off.
- Buttons enable the user to always/never see edge weights, or see them upon mouseover.
- Allows user to set normalization factor so that edge thicknesses for different graphs can be rendered on the same scale.
- Changing the gray threshold deemphasizes weaker edges in the visualization.
- Presenting gray edges as dashed lines further deemphasizes weak edges.

5. View

- Multiple viewport sizes available.
- Graph bounding box can be separated from viewport.
- Zooming and scrolling available.

6. Export

- Basic data export to SIF or GraphML or full GRNmap-compatible Excel workbook, with yeast mRNA degradation rates from Neymotin et al. (2014).
- Image export to PNG, SVG, or PDF.

